

Revisiting “The Impact of Unilateral Divorce on Crime”

A replication and reassessment of Cáceres-Delpiano and Giolito (2012) with heterogeneity-robust difference-in-differences estimators

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Abstract

Cáceres-Delpiano and Giolito (2012) find that the staggered adoption of unilateral divorce across U.S. states raised violent crime by roughly nine percent, an effect they attribute to the reform’s impact on the households children grew up in. In this paper, I replicate their two-way fixed effects estimates using FBI Uniform Crime Report data (1965–1996) and U.S. Census microdata, and then reassess the result with tools that were developed after publication: an event-study specification and the heterogeneity-robust DID_{*i*} estimator of de Chaisemartin and d’Haultfœuille (2024). The original finding survives this reassessment. TWFE estimates place the effect at 8–10 percent across control sets, pre-reform placebo estimates are flat, the effect builds gradually as exposed cohorts reach peak offending ages, and the DID_{*i*} estimates track the TWFE results closely, which indicates that negative weighting is not driving the published estimate.

1. Introduction

Between the late 1960s and the early 1980s, nearly every U.S. state adopted unilateral divorce laws, which allowed one spouse to end a marriage without the other’s consent. Economists have studied the effects of these reforms on many outcomes such as marriage rates, women’s labor supply, and bargaining within the household. Cáceres-Delpiano and Giolito (2012), however, ask a different question: if the reform changed the households that children were raised in, did it also change how much crime those children committed once they became adults?

They find that it did. Using state-year crime data from the FBI’s Uniform Crime Reports combined with individual records from the U.S. Census, they estimate that unilateral divorce raised violent crime rates by roughly nine percent. The effect does not appear immediately, but builds over time as the affected cohorts age into their late teens and twenties. Their explanation is an economic one: the reform left many low-income mothers worse off, and their children paid part of the cost, which shows up years later in crime and institutionalization statistics.

In this paper, I replicate their analysis and test whether the result holds up against methods that were not available in 2012.² I first reproduce the paper’s two-way fixed effects (TWFE) estimates for violent crime and institutionalization, then estimate an event study to check the timing of the effect. This matters because

¹This project began as the final empirical assignment for ECON 452 (Applied Econometrics) at Queen’s University, taught by Dr. Sitian Liu. I have since reworked it into a standalone replication study: correcting and rebuilding the figures, restructuring the analysis, and rewriting the text to stand on its own. All remaining errors are mine.

²As of a literature search conducted in July 2026, no published replication of the paper’s crime result exists: none of the 95 works citing the paper on Google Scholar (29 in RePEc’s CitEc index) re-runs the divorce–crime analysis, and no public replication code is available on GitHub.

recent econometric work has shown that TWFE can be badly biased when treatment effects vary across early and late adopters (de Chaisemartin and d'Haultfœuille 2020). To address this, I re-estimate the violent-crime effect with the heterogeneity-robust DID_t estimator of de Chaisemartin and d'Haultfœuille (2024), along with placebo tests on the pre-reform years.

Overall, the original results hold up well. My TWFE estimates place the effect between eight and ten percent, the event study shows no pre-trends and an effect that grows over time, and the DID_t estimates are close enough to the TWFE ones that negative weighting does not appear to be a serious concern in this setting. However, one caveat should be kept in mind: I am replicating a reduced-form relationship, not the mechanism behind it. The economic story remains the authors' hypothesis, but what I can say with confidence is that the headline number itself is robust.

2. Data

2.1 State-year crime panel

The crime data come from the FBI's Uniform Crime Reports (UCR), covering the 50 states and the District of Columbia from 1965 to 1996, which forms a balanced panel of 1,632 state-year observations. The outcome variable is the violent crime rate per 100,000 residents, in logs. The panel also includes the control variables used in the original paper, which fall into three categories: demographic variables (the fraction of the Black population and the age structure of the population, both entered as category dummies, along with the fraction of immigrants and state population), state aggregate variables (log per-capita income, the unemployment rate, log lagged incarceration, and an indicator for the arrival of crack cocaine), and state policy variables (legal abortion, whether fault is considered in property division, and equitable-division rules).

The treatment timing follows Friedberg's (1998) coding of unilateral divorce laws without separation requirements, as reported in Table 1, Column 1 of Cáceres-Delpiano and Giolito (2012); states whose unilateral divorce law requires a period of separation are treated as non-adopting. I record the year each state adopted unilateral divorce and then generate a treatment dummy equal to one from the year after adoption onward. This is a conservative rule, since it codes the adoption year itself as untreated. Under this coding, 882 state-year observations (54.0 percent) are untreated and 750 (46.0 percent) are treated.

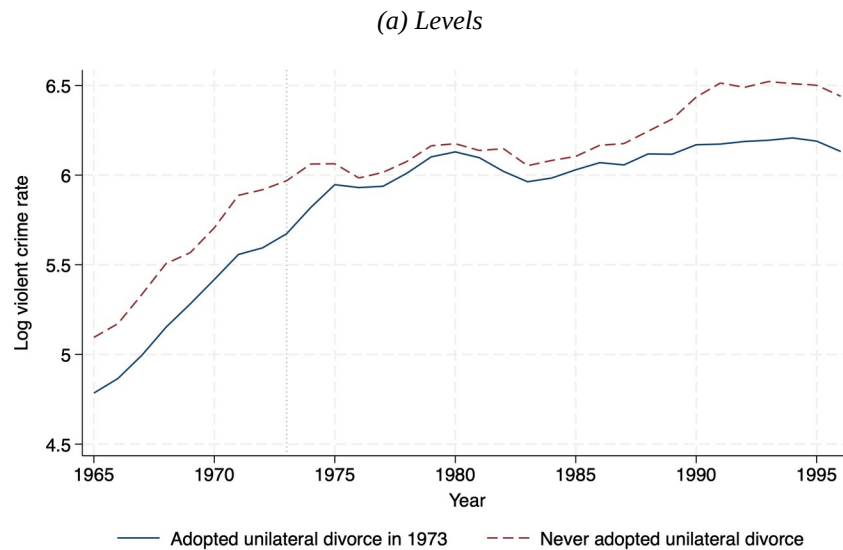
2.2 Census microdata

The individual-level sample comes from IPUMS USA and consists of men aged 15 to 24 in the 1960, 1970, 1980, 1990, and 2000 censuses. Following the original paper, exposure to the reform is assigned by state of birth rather than state of residence, and institutionalization is inferred from group-quarters status. Because the UCR panel does not cover 1960 or 2000, the divorce-law variables for those census years take their 1965 and 1996 values respectively. Divorce law was stable in the surrounding years, so very little is lost from this substitution. The merged sample contains 3,360,455 observations, of which 1,442,569 (42.9 percent) are treated.

3. Motivating evidence

Figure 1 compares the ten states that adopted unilateral divorce in 1973, the most common reform year, with the five states that never adopted unilateral divorce in any form (Arkansas, Delaware, Mississippi, New York, and Tennessee). Panel (a) plots the average log violent crime rate for each group in levels. The control group sits above the treatment group throughout the entire period, largely because it contains New York. However, level differences are absorbed by the state fixed effects in the analysis that follows; what matters for identification is trends, so panel (b) normalizes each group to its 1973 value.

Before the reform, the two series are nearly indistinguishable. After 1973, however, they begin to separate: violent crime in adopting states grows roughly twenty log points faster over the following fifteen years, and the gap only closes again in the early 1990s, when crime in the control states (New York in particular) surged before the national decline set in. This figure is only suggestive, since it rests on sixteen states and includes no controls. But it previews both of the results that the formal analysis delivers: common pre-reform trends, and a positive post-reform gap that emerges gradually rather than immediately.



(b) Normalized to 1973 = 0

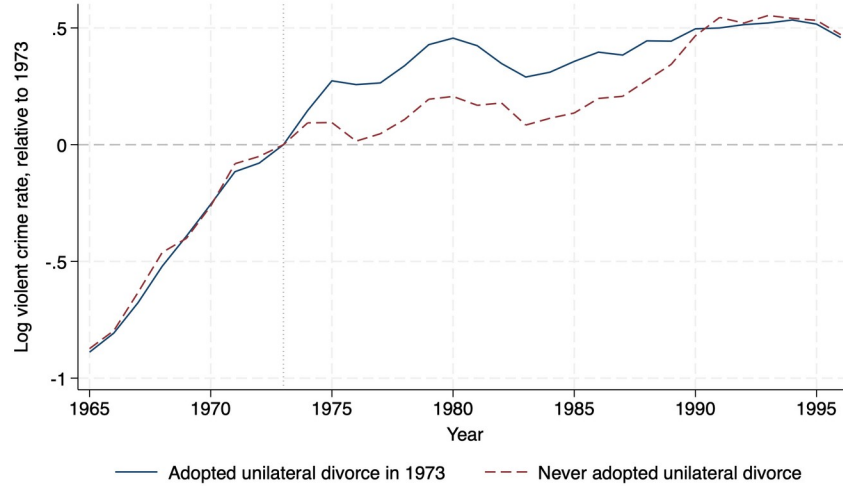


Figure 1. Log violent crime rates, 1965–1996: the ten states adopting unilateral divorce in 1973 versus the five states that never adopted it (AR, DE, MS, NY, TN). Panel (a): group means in levels. Panel (b): each group normalized to its 1973 level. Source: FBI Uniform Crime Reports.

4. Empirical strategy

The baseline specification is the two-way fixed effects model used in the original paper:

$$\ln(\text{violent crime rate})_{st} = \alpha_s + \beta_t + \delta Z_{st} + \gamma U_{st} + \varepsilon_{st}$$

where α_s and β_t are state and year fixed effects, Z_{st} is a vector of time-varying state controls, and U_{st} equals one if state s had adopted unilateral divorce by year t . Standard errors are clustered by state throughout. Under the common trends assumption and homogeneous treatment effects, γ can be interpreted as the average effect of the reform on violent crime.

However, two features of this setting complicate that interpretation. First, common trends is an assumption about a counterfactual, so it cannot be tested directly. The best available evidence is the absence of differential trends before the reform, which the event study examines: replacing the single treatment dummy with a set of indicators for binned years relative to adoption (the bin covering the reform year and the year before is omitted) traces out the full path of the effect and turns the pre-reform bins into placebo tests. Second, adoption is staggered across two decades. de Chaisemartin and d’Haultfœuille (2020) show that with staggered adoption and heterogeneous effects, the TWFE coefficient becomes a weighted average of many two-group comparisons in which already-treated states serve as controls for later adopters, and some of these weights can be negative. To address this, Section 5.4 re-estimates the model with the DID_i estimator of de Chaisemartin and d’Haultfœuille (2024), implemented via `did_multiplegt_dyn` in Stata, which only uses clean comparisons between switching states and not-yet-treated states and therefore remains valid under heterogeneous effects.

5. Results

5.1 Two-way fixed effects estimates

Table 1 reports the TWFE estimates, replicating Panel A of Table 2 in the original paper. With state and year fixed effects alone, adopting unilateral divorce is associated with a 9.7 log point increase in violent crime (SE 4.9). Adding demographic controls lowers the point estimate to 7.2, and adding state aggregate and policy controls moves it to 7.9 and 8.2 respectively. The coefficient stays within a band of 0.07–0.10 in every specification and is significant at the ten-percent level in three of the four; the specification with only demographic controls falls short ($p \approx 0.14$). These results closely match the roughly nine percent effect that the original paper reports.

The stability of the coefficient across columns matters more than any single estimate. Even with state and year fixed effects, there may be time-varying state-level factors that are correlated with both the timing of adoption and crime rates, and the natural candidates, such as demographic composition, economic conditions, and other contemporaneous policy changes, are exactly what the added controls proxy for. The fact that the estimate barely moves when they enter suggests the association is not driven by any one of these confounders, although it cannot rule out factors outside these categories.

	(1)	(2)	(3)	(4)
	Basic	+ Demographic	+ Aggregate	+ Policy
Unilateral divorce	0.0965* (0.0486)	0.0722 (0.0488)	0.0785* (0.0450)	0.0823* (0.0489)
State fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
Demographic controls	No	Yes	Yes	Yes
Aggregate controls	No	No	Yes	Yes
Policy controls	No	No	No	Yes
Observations	1,632	1,632	1,613	1,613

Notes: Dependent variable is the log violent crime rate. UCR state-year panel, 1965–1996. All specifications include state and year fixed effects. Standard errors clustered by state in parentheses. * $p < 0.10$.

5.2 Dynamics and pre-trends

Figure 2 traces the effect in event time. The three pre-reform coefficients are all close to zero and statistically insignificant, which indicates that adopting and non-adopting states were not already diverging before the law changed. This provides supportive evidence for the common trends assumption. However, it is not proof that the assumption holds: the placebo estimates only restrict the observable pre-period, and an unobserved shock that coincided with adoption could still bias the estimates.

The post-reform estimates start near zero and climb steadily, reaching approximately 0.10 eight to fifteen years after the reform and 0.13 in the final bin, more than two decades out. This pattern is informative. A reform that operates through children’s household environments should not move crime immediately, because the exposed cohorts need a decade or more to reach the peak ages for criminal activity. The gradual

build-up is exactly what the mechanism proposed by Cáceres-Delpiano and Giolito predicts, and it is difficult to reconcile with a contemporaneous confounder such as a coincident change in policing.

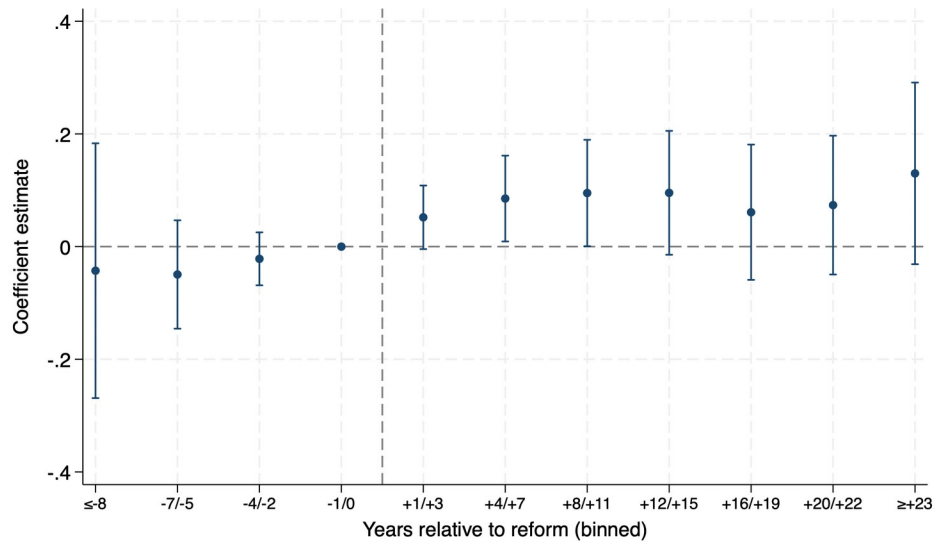


Figure 2. Event-study estimates for the log violent crime rate. Years relative to reform are grouped into bins; the omitted bin ($-1/0$) covers the reform year and the year before, and never-treated states are assigned to it. Vertical bars are 95 percent confidence intervals; standard errors clustered by state.

5.3 Institutionalization in the Census

Table 2 replicates the paper’s individual-level analysis (their Table 7), where the outcome is a dummy equal to one if an individual lives in an institution. For all men aged 15–24, unilateral divorce raises the probability of institutionalization by 0.09 percentage points without additional controls and 0.05 with them; neither estimate approaches statistical significance. For Black men, the point estimates are considerably larger at 0.43 and 0.33 percentage points, although they remain imprecise once standard errors are clustered by state of birth. Given that the mean institutionalization rate for Black men in this sample is approximately 5.1 percent, a 0.43 point increase represents roughly 8 percent of the base rate, which is not trivial.

These results line up with the aggregate findings: outcomes worsen the most where the channel proposed by the original authors, economically vulnerable single-mother households, is most prevalent. However, given the imprecision of the estimates, I read the census evidence as corroborating the UCR results rather than independently establishing them.

	(1)	(2)	(3)	(4)
	All men	All men	Black men	Black men
Unilateral divorce	0.0009 (0.0010)	0.0005 (0.0010)	0.0043 (0.0030)	0.0033 (0.0026)
State-of-birth fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
Age and race dummies	Yes	Yes	Yes	Yes
Age×year, race×year	Yes	Yes	Yes	Yes

State-of-residence FE	No	Yes	No	Yes
Policy controls	No	Yes	No	Yes
Observations	3,360,455	3,360,455	428,920	428,920

Notes: Dependent variable is an indicator for living in an institution (group quarters type 3). Men aged 15–24, U.S. Censuses 1960–2000 (IPUMS). Columns (1)–(2) and (3)–(4) correspond to Columns 1, 2, 5, and 6 of Table 7 in the original paper. Treatment is assigned by state of birth. Standard errors clustered by state of birth in parentheses.

5.4 Heterogeneity-robust estimates

Figure 3 reports the DID_t estimates: dynamic effects for up to ten periods after adoption, along with placebo estimates for five periods before. The placebo estimates are flat and individually insignificant, which is consistent with the event-study evidence on pre-trends. The dynamic effects rise steadily from approximately 0.01 in the first year after adoption to 0.10–0.13 by years six through ten, showing the same magnitude and the same gradual shape as the TWFE event study.

The agreement between the two estimators is itself an important finding. If negative weights were distorting TWFE, the DID_t estimates, which only use clean comparisons against not-yet-treated states, would diverge from it. They do not, which suggests that treatment effects do not vary dramatically across adoption cohorts in this setting. The published TWFE estimate was therefore not an artifact of the estimator, although this is a conclusion that could not have been drawn with the tools available in 2012.

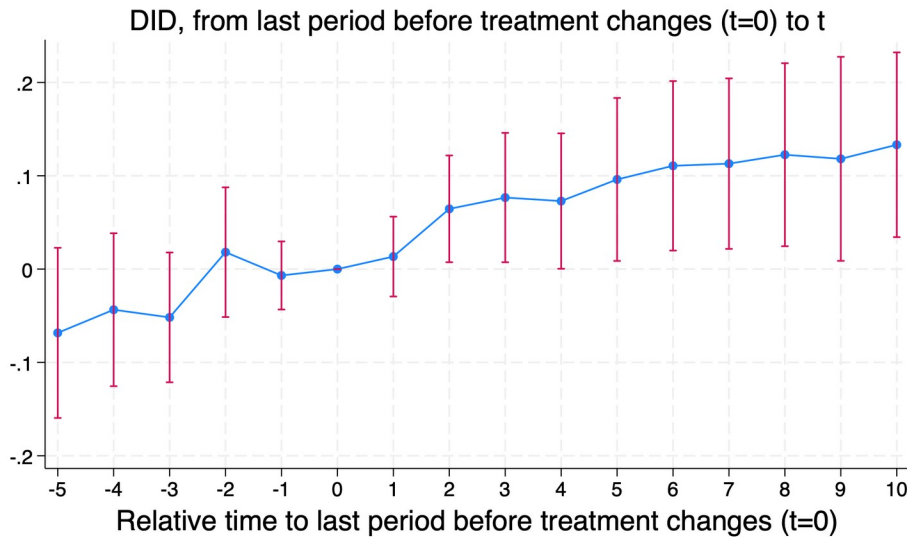


Figure 3. DID_t estimates (de Chaisemartin and d’Haultfœuille 2024) for the log violent crime rate: placebo estimates for five pre-reform periods and dynamic effects up to ten periods after adoption. Vertical bars are 95 percent confidence intervals.

6. Limitations

Three qualifications should be kept in mind. First, the common trends assumption is fundamentally untestable, because we cannot observe what would have happened to crime rates in adopting states had they not adopted unilateral divorce. The flat pre-reform placebos in both the event study and the DID analysis are reassuring, but they only constrain the observed history; a time-varying shock correlated with the timing

of adoption could still bias every estimator used here, since both TWFE and its heterogeneity-robust successors rest on the same assumption.

Second, the outcome variables are imperfect. Aggregate UCR rates reflect reported crime, so changes in reporting or policing that happened to coincide with the reform would contaminate the estimates, and the institutionalization dummy mixes correctional facilities with other types of group quarters. These are the same measurement limitations that the original paper faces, and nothing in the replication resolves them.

Third, nothing in this paper tests the mechanism. The original authors attribute the effect to the deterioration of economic conditions for low-educated single mothers; my analysis neither confirms nor rejects that channel, and the reduced-form estimates are consistent with other stories as well. The broader divorce-law literature also cuts in both directions, since Stevenson and Wolfers (2006) find that the same reforms reduced female suicide and domestic violence. So even if the crime effect is real, it does not settle the overall welfare question.

7. Conclusion

This paper set out to reproduce Cáceres-Delpiano and Giolito's (2012) finding that unilateral divorce raised violent crime, and to test whether that finding survives estimators that were designed after the paper was published. It does. The TWFE estimates place the effect at eight to ten percent and match the original within sampling error. The event study shows no pre-trends and an effect that builds over roughly a decade, which is consistent with a mechanism operating through childhood exposure. Finally, the DID_i estimates track the TWFE numbers closely, which means the negative-weighting problem that has overturned other staggered-adoption results does not overturn this one.

The mechanism itself remains an open question. Establishing whether the effect truly runs through household economics would require data linking family structure, income, and criminal outcomes at the individual level, which is a harder design than a state panel can support. What this replication establishes is narrower but solid: the relationship between unilateral divorce and violent crime documented in the original paper is not an artifact of the estimator that found it.

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